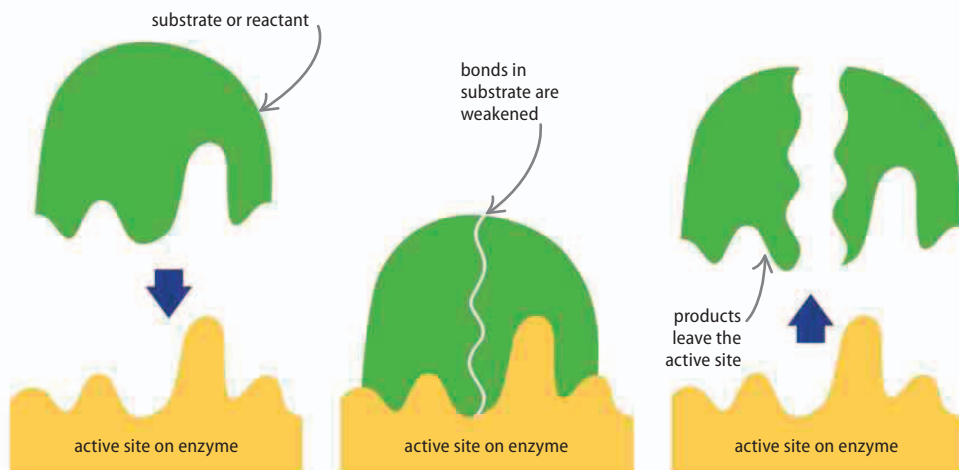


## Enzymes

Most of the chemical reactions that take place inside living bodies would not happen without the catalytic effect of enzymes. Enzymes are protein molecules that are highly folded into shapes specific to their roles. These shapes create an area known as the enzyme's active site. The reactants—known as substrates in biochemistry—are also molecules with complex shapes. They fit onto the enzyme's active site, where the reaction takes place. Enzymes are used in the digestive system to break down large molecules of food into smaller ones.



### △ Active site

Only a specific substrate can bond to a specific enzyme's active site, like a key fitting into a lock.

### △ Catalyzed reaction

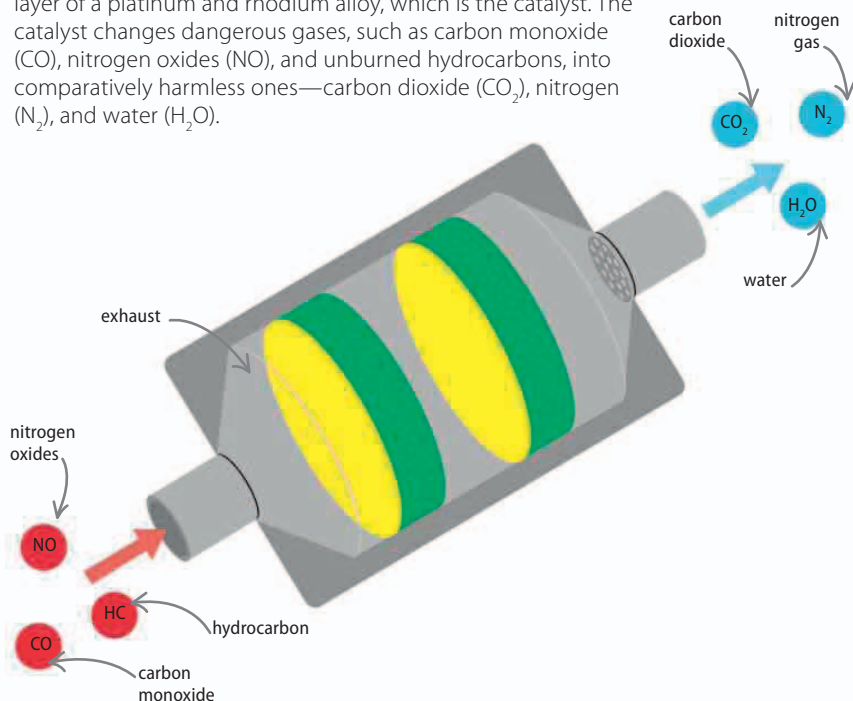
When joined to the enzyme, the chemical reaction can take place. Bonds within the substrate are weakened.

### △ Products produced

The substrate divides into two products. These break off the active site, leaving it free to collect a new substrate.

## Catalytic converter

Every new car is fitted with a catalytic converter, or "cat." The engine exhaust passes through this device before it enters the air. Inside is a honeycomb-shaped ceramic coated with a thin layer of a platinum and rhodium alloy, which is the catalyst. The catalyst changes dangerous gases, such as carbon monoxide (CO), nitrogen oxides (NO), and unburned hydrocarbons, into comparatively harmless ones—carbon dioxide (CO<sub>2</sub>), nitrogen (N<sub>2</sub>), and water (H<sub>2</sub>O).



### REAL WORLD

#### Margarine

Margarine is made using a catalyst. The starting materials are vegetable oils: long chain molecules made from carbon and hydrogen. The oils are unsaturated—their molecules have room for more hydrogen atoms. Hydrogen is bubbled through the oil over a nickel catalyst, which saturates the oil molecules, turning them into a butterlike solid.

